MVP — Personal Item Inventory

1. Problem

1.1. Problem statement

People who own durable items that are used infrequently often rely on memory or informal lists to keep track of what they own and where those items are stored. As a result, they may purchase items they already own, spend significant time searching for rarely used belongings, or keep items for years simply because they are unsure whether they may need them in the future. These behaviors can contribute to unnecessary spending, wasted time, and accumulation of unused possessions

1.2. Evidence

Five people were interviewed, and the following questions were asked:

- How often does this happen?
- What types of objects are involved?
- How much time do you spend looking for things?
- Do you maintain any kind of list?
- Have you ever thrown something away because you didn't know whether you might need it later?
- Have you kept something for years because you're afraid you'll need it someday?

1.3. Key findings

The main findings are:

Evidence area	Findings
Duplicate purchases	4 out of 5 people reported having bought something they already owned at least once.
Frequency	In general, it does not occur very frequently; responses range from occasionally to approximately once a year.
Types of objects	The items mainly include small or low-cost items: food/snacks, drinks, stationery, books, tools, electronics, kitchen items, spare parts, clothing.
Searching for objects	The time it takes varies considerably. Some people hardly look at all because they remember where they put things; others report taking anywhere from 30 minutes to over an hour in certain situations.
Lists / tracking	There are mixed behaviors: some people use lists, notes or memory; others do not maintain any inventory.
Discarding items	Several people have thrown away items and later regretted it or wondered whether they should have kept them.
Keeping “just in case”	4 out of 5 show some behavior of keeping things because they might need them later, although they are not necessarily frequently used objects.
Location problem	The problem is stronger for items that are used infrequently and stored away, rather than everyday objects.

1.4. Workaround

Currently, users employ the following solutions to the problem at hand:

- Search closets, garage, storage rooms.
- Keep lists in notebooks.
- Rely on memory.
- Ask another family member.
- Simply buy another item when they can't find the first one.

2. Target User

2.1. Primary target user

The primary target users are people who own a significant number of durable items that they use infrequently and rely mainly on memory or informal lists to keep track of them. They may own tools, electronics, spare parts, games, household items, or other belongings that are stored away and are not part of their daily routine.

2.2. Possible segments

- Tool/equipment owners: tools, spare parts, electronics, equipment.
- Hobbyists: board games, art supliers, sports equipment, DIY materials, electronics
- General household users

3. Goals

The MVP should help users:

- Find an item when they need it.
- Remember whether they own an item and where it is stored.
- Decide whether to keep, discard, sell, or donate an item.
- Avoid unnecessary purchase.

4. Value Proposition

These goals are supported by four main value areas:

4.1. Ownership Memory: Avoid duplicate purchases

Help users remember what they own so they can check whether they already have an item before buying another one.

4.2. Location & Retrieval: Find rarely used items quickly

Help users remember where an item is stored and retrieve it without spending unnecessary time searching.

4.3. Decision Support: Make better decisions about what to do with an item

Help users decide whether to keep, discard, sell, or donate belongings based on relevant information.

4.4. Low-Maintenance Inventory: Track only what is difficult or costly to manage

The system does not require users to create a complete inventory. Users can choose which belongings are worth registering.

Key hypothesis

Users will perceive enough value from tracking selected belongings to justify the effort required to maintain the inventory.

5. MVP Features

5.1. Add / Register Item

The user can register a selected belonging.
Minimum information

- Item name
- Category
- Storage location level 1
- Storage location level 2

Optional information

- Photo
- Condition
- How often used
- Quantity
- Unit of Measure

The user decides which belongings are worth registering. The system should keep item registration simple and minimize the amount of information required.

5.2. Storage Location

The user can assign a two-level storage location to an item.

Example:

Home → Garage

Level 1 and Level 2 locations are managed by the user through Settings.

5.3. Search Inventory

The user can search for an item by name.

Example:

"Hammer"

Result:

Home → Garage

The system returns matching registered items and their storage locations.

If no matching item is found, the system should clearly indicate that no matching item was found in the user's inventory.

5.4. Item Details

The user can view the information associated with an item.

Example:

- Cordless Drill
- Category: Tools
- Location: Home → Garage
- Condition: Good
- How often used: Rarely
- Quantity: 1
- Unit of Measure: pcs
- Decision: Keep

The exact fields should remain minimal and only include information relevant to the MVP.

5.5. Item Status / Decision

The user can assign one of the following decisions to an item:

- Keep
- Sell
- Donate
- Discard

The system helps the user review relevant information and record their decision.

The system does not recommend which decision the user should make.

5.6. Decision-Relevant Information

The system allows the user to store a small amount of information that may support their decision about an item.

MVP information includes:

- Condition
- How often it is used
- Quantity
- Unit of Measure

Additional information such as replacement cost or last used may be considered later if validation shows that it provides value.

5.7. Edit Item

The user can update item information when something changes.

For example:

Home → Garage TO Home → Storage Room

The user can update information such as the storage location, condition, quantity, or decision.

5.8. Configuration Management

The user can maintain the configuration values used when creating or editing items.

Categories

The user can:

- Create a category
- Edit a category
- Activate or deactivate a category

Inactive categories remain in the system but are not normally available for new item selection.

Storage Locations

The user can:

- Create Level 1 locations
- Edit Level 1 locations
- Activate or deactivate Level 1 locations
- Create Level 2 locations under a Level 1 location
- Edit Level 2 locations

- Activate or deactivate Level 2 locations

Inactive storage locations remain in the system but are not normally available for new item selection.

Units of Measure

The system provides a predefined list of common units of measure.

Predefined units cannot be modified by the user.

The user can also:

- Create a unit of measure
- Edit a unit of measure

Custom units are user-managed and can be modified after creation.

5.9. System-Defined Values

The following values are predefined by the system and cannot be modified by the user:

How Often Used

- Daily
- Weekly
- Monthly
- Occasionally
- Rarely
- Never

Condition

- New
- Good
- Fair
- Poor
- Broken

The exact predefined values should be validated during prototyping.

6. MVP Scope

The MVP will provide a simple personal inventory that helps users remember what they own, find rarely used belongings, and record decisions about what to do with them.

Users decide which belongings are worth registering. The system does not require a complete inventory.

6.1. Inventory Management

- Add / register an item
- Edit an item
- Delete an item
- View item details

6.2. Ownership Memory

- Store information about registered belongings
- Search the inventory by item name
- Check whether an item is already registered

6.3. Location & Retrieval

- Assign a two-level storage location to an item
- View the item's storage location
- Search for an item and retrieve its location

Storage locations follow the structure:

Level 1 → Level 2

Example:

Home → Garage

6.4. Decision Support

- Review relevant information about an item
- Record one of four decisions:
    - Keep
    - Sell
    - Donate
    - Discard

The system does not recommend a decision. The user makes the decision, and the system stores it.

6.5. Configuration Management

Users can maintain the configuration values used when creating or editing items.

Categories

- Create categories
- Edit categories
- Activate / deactivate categories

Storage Locations

- Create and edit Level 1 locations
- Create and edit Level 2 locations
- Activate / deactivate locations

Units of Measure

- Use predefined common units of measure
- Create custom units of measure
- Edit custom units of measure
  Predefined units of measure cannot be modified by the user.

6.6. Low-Maintenance Inventory

- Keep item registration simple
- Minimize the amount of information required
- Make item updates easy
- Allow users to track only selected belongings

6.7. System-Defined Values

The following values are predefined by the system and cannot be modified by the user:

- How Often Used
- Condition
- Item Decision values: Keep / Sell / Donate / Discard

6.8. Out of Scope

- Automatic duplicate detection
- AI-generated recommendations
- Marketplace integration
- Automatic selling or donation
- Social or sharing features
- Household collaboration
- Storage hierarchies with more than two levels
- Automatic selection of belongings to track

7. Functional Requirements

7.1. Inventory Management

FR-01 — Add Item

The system shall allow the user to create an item record.

The item shall contain, at minimum:

- Name
- Category
- Storage location Level 1
- Storage location Level 2

Optional information may include:

- Photo

- Condition
- How often used
- Quantity
- Unit of Measure

FR-02 — View Item

The system shall allow the user to view the information associated with a registered item.

FR-03 — Edit Item

The system shall allow the user to modify an item's information, including its category, storage location, condition, usage frequency, quantity, unit of measure, and decision.

FR-04 — Delete Item

The system shall allow the user to remove an item from the inventory.

7.2. Ownership Memory

FR-05 — Search Item

The system shall allow the user to search the inventory by item name.

FR-06 — Identify Ownership

The system shall allow the user to determine whether a matching item exists in the inventory by searching for the item's name.

The system should only indicate whether a matching registered item was found; no result does not mean that the user does not own the item.

7.3. Location & Retrieval

FR-07 — Store Location

The system shall allow the user to associate an item with a two-level storage location:

Level 1 → Level 2

Level 2 locations shall belong to a Level 1 location.

FR-08 — Display Location

The system shall display the item's Level 1 and Level 2 storage location when the user views the item.

7.4. Decision Support

FR-09 — Record Item Decision

The system shall allow the user to record one of the following decisions for an item:

- Keep
- Sell
- Donate
- Discard

The system shall not recommend a decision.

FR-10 — Store Decision-Relevant Information

The system shall allow the user to store a limited amount of information that may support their decision about an item.

MVP information includes:

- Condition
- How often used
- Quantity
- Unit of Measure

Additional information such as approximate replacement cost or last used may be considered later if validation shows that it provides value.

7.5. Configuration Management

FR-11 — Manage Categories

The system shall allow the user to:

- Create categories
- Edit categories
- Activate categories
- Deactivate categories

Inactive categories shall not be available for new item selection but may remain associated with existing items.

When a category used by existing items is deactivated, the system identifies the affected items and notifies the user that their information requires an update.

FR-12 — Manage Storage Locations

The system shall allow the user to:

- Create Level 1 storage locations
- Edit Level 1 storage locations

- Activate Level 1 storage locations
- Deactivate Level 1 storage locations
- Create Level 2 storage locations under a Level 1 location
- Edit Level 2 storage locations
- Activate Level 2 storage locations
- Deactivate Level 2 storage locations

Inactive storage locations shall not be available for new item selection but may remain associated with existing items.

A Level 2 location belongs to one Level 1 location.

Deactivating a Level 1 location automatically deactivates its Level 2 locations.

FR-13 — Manage Units of Measure

The system shall provide a predefined list of common units of measure.

Predefined units of measure shall not be editable by the user.

The system shall allow the user to create custom units of measure.

The system shall allow the user to edit custom units of measure.

8. Acceptance Criteria

FR-01 — Add Item

Given the user has entered the required item information, including item name, category, storage location Level 1, and storage location Level 2, when the user submits the item, then the system creates and stores the item successfully.

The system should prevent saving when any required information is missing.

Optional information may be omitted.

FR-02 — View Item

Given a registered item exists, when the user opens the item, then the system displays the item's stored information.

FR-03 — Edit Item

Given an existing item, when the user changes its information and saves it, then the updated information is stored and displayed.

FR-04 — Delete Item

Given an existing item, when the user deletes the item, then the item is removed from the inventory and no longer appears in the inventory or search results.

FR-05 — Search Item

Given an item exists in the inventory, when the user searches using its name, then the matching item appears in the search results.

The system should provide understandable feedback when no matching item is found.

FR-06 — Identify Ownership

Given the user is considering purchasing an item, when they search the inventory using the item's name, then they can determine whether a matching registered item exists.

No search result should only indicate that no matching registered item was found.

FR-07 — Store Location

Given an item is being created or edited, when the user selects an active Level 1 location and an active Level 2 location belonging to that Level 1 location, then the system stores the selected location with the item.

FR-08 — Display Location

Given an item has a stored Level 1 and Level 2 location, when the user opens the item, then the system displays the storage location clearly as:

Level 1 → Level 2

FR-09 — Record Item Decision

Given the user is viewing an item, when they select and save a decision, then the system stores and displays the selected decision.

Valid decisions are:

Keep / Sell / Donate / Discard

The system should not recommend a decision.

FR-10 — Store Decision-Relevant Information

Given the user is creating or editing an item, when they enter decision-relevant information and save the item, then the system stores and displays the entered information.

The available MVP information includes:

- Condition

- How often used
- Quantity
- Unit of Measure

FR-11 — Manage Categories

Given the user is in Settings, when the user creates a category and saves it, then the category becomes available for item selection.

Given an existing category, when the user edits and saves it, then the updated category information is displayed.

Given an active category, when the user deactivates it, then the category is no longer available for new item selection.

Given an inactive category, when the user activates it, then the category becomes available for new item selection.

Given an active category is associated with existing items, when the user deactivates the category, then the category becomes inactive and the system identifies the affected items as requiring an information update.

FR-12 — Manage Storage Locations

Given the user is in Settings, when the user creates a Level 1 location and saves it, then the location becomes available for item selection.

Given an active Level 1 location, when the user creates a Level 2 location under it and saves it, then the Level 2 location becomes available for item selection.

Given an existing location, when the user edits and saves it, then the updated location information is displayed.

Given an active storage location, when the user deactivates it, then the location is no longer available for new item selection.

Given an inactive storage location, when the user activates it, then the location becomes available for new item selection.

Given an active storage location is associated with existing items, when the user deactivates the location, then the location becomes inactive and the system identifies the affected items as requiring an information update.

Given an active Level 1 location has active Level 2 locations, when the user deactivates the Level 1 location, then the Level 1 location and all associated Level 2 locations become inactive.

FR-13 — Manage Units of Measure

Given the user is entering or editing an item quantity, when the user opens the Unit of Measure selector, then the system displays the predefined units of measure and any custom units created by the user.

Given a predefined unit of measure exists, when the user views the unit in Settings, then the unit cannot be edited.

Given the user creates a custom unit of measure and saves it, then the custom unit becomes available for item selection.

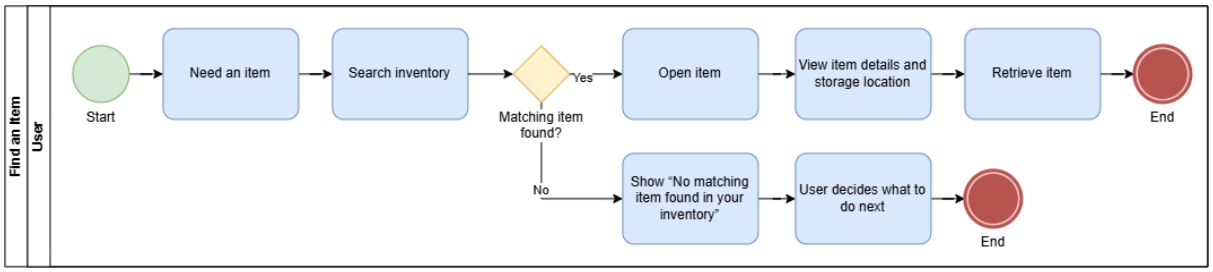
Given an existing custom unit of measure, when the user edits and saves it, then the updated unit information is displayed and available for item selection.

9. User Flow / UX

The UX should focus on the main user situations identified in the problem and interviews, while keeping the main interactions simple and low-maintenance.

9.1. Main Flow — Find an Item

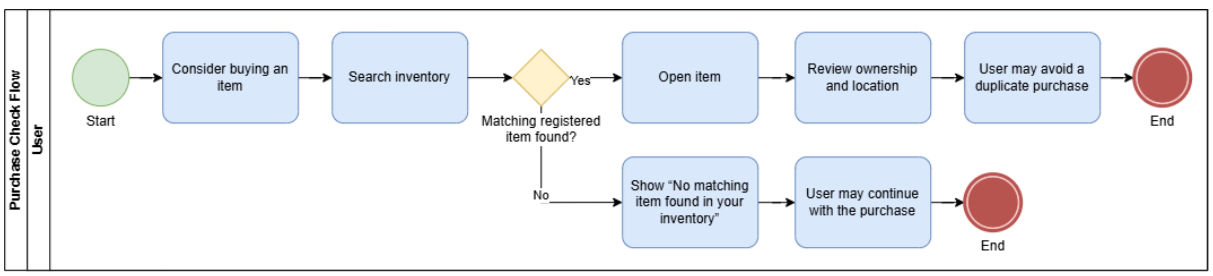
Goal: Find a registered item and retrieve its stored location



The system should make the item's location easy to understand and avoid unnecessary navigation.

9.2. Purchase Check Flow

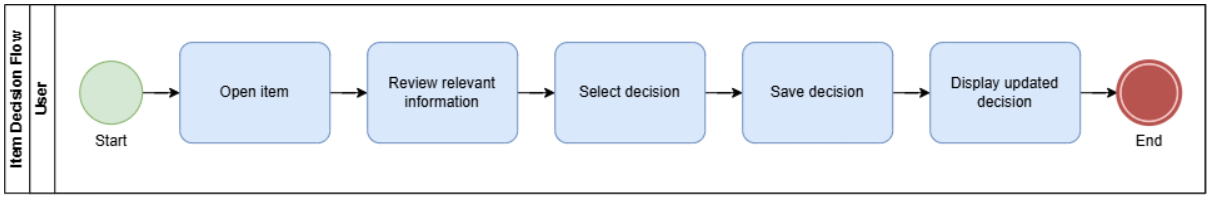
Goal: Check whether the user already owns an item before purchasing it.



No search result means that no matching registered item was found. It does not mean that the user does not own the item.

9.3. Item Decision Flow

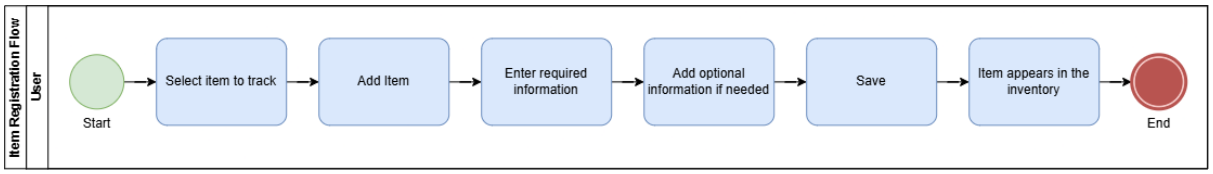
Goal: Review an item and record a decision.



The system does not recommend a decision. The user makes the decision while the system stores and presents the relevant information.

9.4. Item Registration Flow

Goal: Register a selected belonging with a small amount of information.

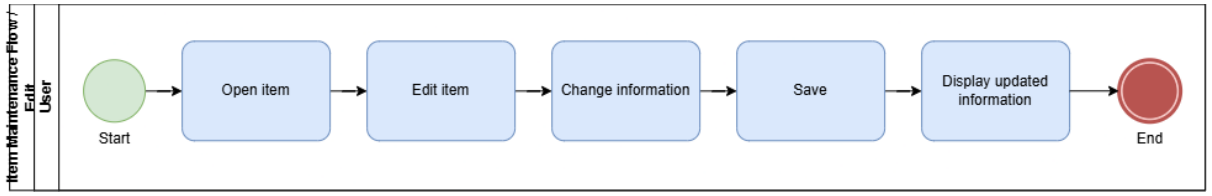


The user decides which belongings are worth registering. The system does not require a complete inventory.

9.5. Item Maintenance Flow

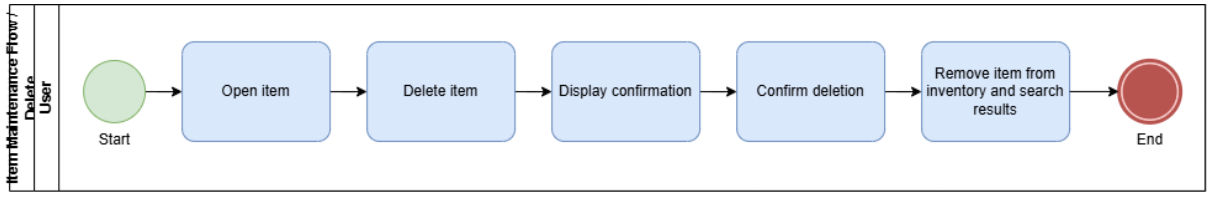
Goal: Keep item information reasonably current.

Edit



Examples include changing the category, storage location, condition, quantity, unit of measure, or decision.

Delete



9.6. Configuration Management Flow

Goal: Maintain the configuration values used when creating or editing items.

Inventory → Settings → Select configuration type

Available configuration types:

- Categories
- Storage Locations
- Units of Measure

Categories and storage locations can be created, edited, activated, and deactivated.

Units of measure include predefined system values and user-created custom values. Predefined units cannot be modified, while custom units can be created and edited.

9.7. Category Management Flow

Goal: Maintain available item categories.

Settings → Categories → Create / Edit / Activate / Deactivate category → Save → Updated category list is displayed

Inactive categories are not normally available for new item selection.

If a category used by existing items is deactivated, the system identifies the affected items and notifies the user that their information requires an update.

9.8. Storage Location Management Flow

Goal: Maintain two-level storage locations.

Settings → Storage Locations → Select Level 1 → View Level 2 locations → Create / Edit / Activate / Deactivate location

The location structure is:

Level 1 → Level 2

When a Level 1 location is deactivated, all associated Level 2 locations are automatically deactivated.

Inactive locations are not normally available for new item selection.

If a storage location used by existing items is deactivated, the system identifies the affected items and notifies the user that their information requires an update.

9.9. Unit of Measure Management Flow

Goal: Maintain units of measure used when entering item quantities.

Settings → Units of Measure → View predefined and custom units

Predefined Units: Predefined units are provided by the system and cannot be modified.

Custom Units:

Create custom unit → Save → Unit becomes available for item selection

Edit custom unit → Save → Updated unit is displayed and available for item selection

9.10. Information Update Flow

Goal: Help users update items that reference inactive configuration values.

Configuration value becomes inactive

- System identifies affected items
- User receives notification
- User reviews affected items
- User updates the item information
- Item no longer requires an update

The system should not automatically change the item's information. The user decides which valid category or storage location should replace the inactive value.

9.11. Important UX States

The main flows should account for:

- Empty inventory
- No search results
- One matching result
- Multiple matching results
- Item without optional information
- Item without a decision
- Editing an existing item
- Delete confirmation
- Successful save
- Missing required information
- Save or system error
- Inactive category
- Inactive Level 1 storage location
- Inactive Level 2 storage location
- Existing item referencing an inactive category or storage location
- Items requiring an information update
- Predefined unit of measure
- Custom unit of measure

The UX should make the next action clear in each state and provide understandable feedback when an action cannot be completed.

10. Risks & Open Questions

10.1. Risk 1 — Maintenance Effort

Risk

Users may find registration and maintenance more effortful than the value they receive.

Fallback

Minimize required fields, keep the basic item record simple, and make updates quick.

10.2. Risk 2 — Data Accuracy

Risk

Storage locations, item information, or decisions may become outdated, reducing the usefulness of the inventory.

Fallback

Make editing quick and visible so users can update information when something changes.

10.3. Risk 3 — Decision Support Scope

Risk

The decision-support feature could expand into AI recommendations, pricing, resale integration, or complex scoring.

Fallback

For the MVP, the user makes the decision. The system only stores and presents relevant information and the selected decision.

10.4. Risk 4 — Storage Model Complexity

Risk

Managing relationships between Level 1 and Level 2 storage locations could become unnecessarily complex.

Fallback

Limit the MVP to exactly two storage levels and keep the relationship simple:

Level 1 → Level 2

10.5. Risk 5 — Low Frequency of the Problem

Risk

The problem does not occur very frequently for many users. Duplicate purchases were reported as occasional events, in some cases approximately once a year.

Users may therefore not perceive enough value from maintaining an inventory to justify the effort.

Fallback

Keep the MVP low-maintenance and focus on selected belongings that are difficult to remember, rarely used, or time-consuming to find. Validate whether the perceived benefit is sufficient to justify the maintenance effort.

10.6. Risk 6 — Limited Evidence for Decision Support

Risk

The interviews provide evidence of keeping items “just in case” and regretting some discarded items, but they do not yet establish that users need a dedicated decision-support feature.

Fallback

Keep the feature simple: allow users to review a small amount of relevant information and record Keep / Sell / Donate / Discard. Validate whether users find this useful.

10.7. Risk 7 — Configuration Management

Risk

Categories and storage locations may become inconsistent as users create, edit, activate, and deactivate them, while custom units of measure may become inconsistent as users create and modify them.

Fallback

Keep configuration management simple and clearly distinguish between system-defined values and user-defined values.

Predefined units of measure remain controlled by the system, while custom units can be edited by the user.

10.8. Open Questions

- How much information are users willing to enter when registering an item?
- Is avoiding occasional duplicate purchases valuable enough to justify maintaining the inventory?
- How much time are users willing to spend registering and updating an item?
- Which information is actually useful when deciding Keep / Sell / Donate / Discard?

- How should the system handle similar item names, such as multiple versions of the same type of item?
- Will users continue maintaining the inventory over time?
- Does the proposed low-maintenance approach provide enough value without requiring users to track everything they own?

11. Success Criteria

The MVP should be considered successful if it demonstrates value across the four proposed value propositions while supporting the required configuration capabilities.

11.1. Ownership Memory

Users can use the inventory to answer:

“Do I already own this?”

Evidence of success:

- Users can find registered items through search.
- Users can understand whether a matching item exists.
- Users report that this information can help them avoid unnecessary purchases.

11.2. Location & Retrieval

Users can answer:

“Where is it?”

Evidence of success:

- Users can find an item's stored Level 1 and Level 2 location.
- The location information is understandable.
- Users can complete the retrieval flow without unnecessary confusion.

11.3. Decision Support

Users can review information and record a decision about an item.

Evidence of success:

- Users can review relevant information.
- Users can assign Keep / Sell / Donate / Discard.
- Users consider the stored information useful when making the decision.

11.4. Configuration Management

Users can maintain the configuration values required by the inventory.

Evidence of success:

- Users can create and edit categories and storage locations.
- Users can activate and deactivate categories and storage locations.
- Inactive categories and storage locations are not normally available for new item selection.
- The system identifies existing items that reference inactive categories or storage locations.
- The system notifies the user when an item requires an information update.
- Deactivating a Level 1 storage location automatically deactivates its Level 2 locations.
- The system provides a predefined list of common units of measure.
- Predefined units of measure cannot be modified by the user.

11.5. Low-Maintenance Inventory

Users can maintain their inventory without excessive effort.

Evidence of success:

- Users can add an item with only a small amount of information.
- Users can update information easily.
- Users do not need to create a complete inventory to obtain value.
- Users perceive the maintenance effort as reasonable compared with the benefit.